

INFORMATION TECHNOLOGY 110 LAB A LAB ASSIGNMENT 1

Due: Jan 22nd 11:59 pm.

Topics:

Base Conversions

Part 1: Binary Numbers

- a. Take the six digits of your student ID number and split the number into two 3-digit numbers, named n1 ($d_1 d_2 d_3$) and n2 (d_4, d_5, d_6). Convert the two numbers n1 and n2 into their equivalent binary numbers. Show your work when doing the conversions.
- b. Convert your six-digit student ID number as follows: Replace each digit with 1 if it is odd and with 0 if it is even. For example, if $n = 123456$, then $b = 101010$. Add a 1 to the beginning of the number. For example, b becomes 1101010. Treat the final number as binary. Convert it into decimal. Show your work when doing the conversion.

Part 2: Hexadecimal Numbers

- a. Convert your student ID number into its equivalent hexadecimal number. Show your work when doing the conversions.
- b. Convert the hexadecimal number from #2a back into its equivalent decimal number. Show your work when doing the conversions.

Submit the work of all questions for grading to Moodle.